

Workforce Insights: Employee Attrition & HR Analytics Dashboard

1. Executive Summary

This project focuses on analyzing employee workforce data to identify key factors influencing employee attrition, workforce demographics, job satisfaction, and organizational performance. The dashboard was developed in Microsoft Power BI by transforming raw HR data into meaningful business insights using Power Query, DAX measures, and interactive visualizations.

The objective of this project is to help HR professionals and business leaders monitor workforce trends, understand employee turnover patterns, identify high-risk employee groups, and support data-driven workforce planning. The dashboard enables users to filter information dynamically across departments, job roles, age groups, education fields, and other employee attributes, making HR decision-making faster and more effective.

2. Project Objectives

- Analyze employee attrition trends.
 - Measure workforce demographics.
 - Identify departments with high employee turnover.
 - Analyze employee satisfaction levels.
 - Understand the relationship between overtime and attrition.
 - Evaluate employee income distribution.
 - Monitor workforce experience and tenure.
 - Support strategic HR decision-making.
-

3. Business Problem

Employee attrition is one of the major challenges faced by organizations. High employee turnover increases recruitment costs, affects productivity, and impacts overall business performance. HR departments often struggle to identify the underlying reasons behind employee resignations due to scattered and unorganized workforce data.

This project aims to transform raw employee records into an interactive HR analytics dashboard that provides meaningful insights into employee demographics, satisfaction, compensation, experience, and attrition. These insights help HR managers identify

workforce trends, improve employee retention strategies, and make informed business decisions.

4. Dataset Overview

Dataset Name

HR Analytics Dataset

Number of Records

- 1480 Employees

Number of Columns

- 38 Variables

Major Features

- Employee ID
 - Age
 - Gender
 - Department
 - Job Role
 - Monthly Income
 - Education
 - Marital Status
 - Overtime
 - Business Travel
 - Job Satisfaction
 - Performance Rating
 - Work-Life Balance
 - Years at Company
 - Attrition
 - Salary Slab
-

5. Data Cleaning & Preprocessing

Mention the work you performed.

- Imported CSV into Power BI.
- Checked missing values.
- Verified data types.
- Removed duplicate records (if any).
- Standardized categorical values.
- Created calculated columns.
- Created DAX measures.
- Optimized relationships.
- Prepared data model.
- Built interactive filters.

6. Tools & Technologies

Tool	Purpose
Power BI	Dashboard Development
Power Query	Data Cleaning
DAX	KPI Calculations
CSV Dataset	Data Source
Excel	Initial Inspection

7. Data Modeling

Explain:

- Imported HR dataset.
- Cleaned data using Power Query.
- Created calculated measures using DAX.
- Built star-like analytical model.